

Let T be an operator on a fdim. v. space V over a field F

A nonzero vector v is called an eigenvector for T if $T(v) = \lambda v$ for $\lambda \in F$

The scalar λ is called an eigenvalue of T corresponding to v ($v = \text{"}\lambda\text{-evector of } T\text{"}$)

Short: evector, evaluate ① We first find evals of T as follows:

T is diagonalizable

$\Leftrightarrow \exists$ a basis of V wrt which matrix of T is diagonal

$\Leftrightarrow V$ has a basis consisting of evector of T .

Name: Eigenbasis of V
Corresp. to T

$$Tv = \lambda v \Leftrightarrow \lambda \text{Id}(v) - T(v) = 0 \Leftrightarrow (\lambda I - T)v = 0 \Leftrightarrow v \in \ker(\lambda I - T)$$

! λ -eigenspace $\ker(\lambda I - T) \neq 0 \Leftrightarrow \boxed{\det(\lambda I - T) = 0}$ "char eq" " λ -eigenspace of T "

Evals of T are solutions of this eq

② For each eval λ , solve $(\lambda I - T)v = 0$ for v using row reduction and get a basis

If $T(v) = Av$ for a square matrix A with $V = F^n$ then

" A is diagonalizable" $\iff \exists$ invertible P s.t. $\left\{ \begin{array}{l} P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_n) = D \end{array} \right.$

Observe let cols of $P = [v_1, \dots, v_n]$

Then $v_1, \dots, v_n$ is an eigenbasis for $T(v) = Av \rightsquigarrow$

$$\left\{ \begin{array}{l} \text{ie. } AP = PD \\ \text{ie. } A[v_1 \dots v_n] = [v_1 \dots v_n] \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \\ \text{ie. } [Av_1 \dots Av_n] = [\lambda_1 v_1 \dots \lambda_n v_n] \end{array} \right.$$

Examples ① $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \rightsquigarrow \lambda = \frac{1 \pm \sqrt{5}}{2} \rightsquigarrow$ get ebasis v_1, v_2
char eqⁿ $\lambda^2 - \lambda - 1 = 0$ Get Fibonacci formula.

cols of AP

② $R = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \rightsquigarrow \lambda^2 + 1 = 0$ no roots in $\mathbb{R}$ but think of $T(v) = Rv$ for $v \in \mathbb{C}^2$. Non diagonalize T !

$$R \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} -i \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ i \end{bmatrix} \text{ evector with eval } -i$$

Defⁿ A matrix (or an operator) is called nilpotent if some power of it is 0 .

③ $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \rightsquigarrow \lambda^2 - 1 = 0$ e basis $\begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$
 e value $1, -1$

④ Exercises $A = \begin{bmatrix} 2 & 2 \\ 1 & 3 \end{bmatrix}$ $R = \begin{bmatrix} c & s \\ -s & c \end{bmatrix}$

$$B = \begin{bmatrix} -2 & 0 & 1 \\ -5 & 3 & 1 \\ 4 & -2 & -1 \end{bmatrix}$$

$$\left. \begin{array}{l} c = \cos \theta \\ s = \sin \theta \end{array} \right\} \theta \neq 0, \pi$$

$$U = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix} \quad \lambda I - U = \begin{bmatrix} \lambda - 1 & -4 & -5 \\ 0 & \lambda - 2 & -6 \\ 0 & 0 & \lambda - 3 \end{bmatrix}$$

$$\text{Char poly} = (\lambda - 1)(\lambda - 2)(\lambda - 3)$$

$$U' = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 1 & 6 \\ 0 & 0 & 3 \end{bmatrix} \rightsquigarrow \text{e values} = 1, 3$$

Is U' diagonalizable? Find out!

Ex no nonzero nilpotent matrix is diagonalizable

⑤ $N = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is not diagonalizable.

0 is the only e value. $\Rightarrow$ If N were diagonalizable then $P^{-1}NP = \text{diag}(0, 0)$

$$\Rightarrow N = 0 \rightarrow \leftarrow = 0$$

Ex. Let $v_1, \dots, v_k$ be eigenvectors of T with k distinct values, $\lambda_1, \dots, \lambda_k$ resp. Show that $v_1, \dots, v_k$ are linearly indep.

Induction on k

or of exercise
Let $\dim V = n$. Let T be an operator on V

Consequence: T spaces with distinct values are linearly indep. i.e. sum of eigenspaces of any operator is direct

$\therefore T$ is diagonalizable

Write a formal proof of this

$$\sum \dim \text{of spaces} = \dim V$$

T has n distinct values $\Rightarrow T$ is diagonalizable

(In this case each space is 1-dim)

Converse is utterly false e.g. $T = Id$ or 0 .

Evalues of triangular matrix are just its diagonal entries $a_{11}, \dots, a_{nn}$ b/c char poly = $\prod (\lambda - a_{ii})$